



ORIGINAL RESEARCH ARTICLE

Fabrication of AISI 434L Stainless Steel Thin Wall Structures by TIG-Aided Powder Bed Fusion Arc Additive Manufacturing: Evaluation of Metallurgical Characteristics and Mechanical Properties

M.D. Aseef Khan and Manoj Masanta

Submitted: 19 March 2024 / Revised: 1 July 2024 / Accepted: 7 July 2024 / Published online: 7 August 2024

Thin wall structure with five-layer depositions has been fabricated by TIG-based powder bed fusion-type arc additive manufacturing (TIG PBF-AAM) method using AISI 434L stainless steel powder as feedstock. Suitable processing conditions were identified from the trial experiments of single-layer deposition, and thereafter multilayer depositions were executed to attain a thin wall structure. The effect of the processing conditions on the part geometry has also been investigated. Microstructural analysis revealed that the fabricated structure is dominated with α -ferrite phase along with martensite present mostly at the grain boundaries. Microhardness value in the fabricated part was recorded within a range of 186–330 HV_{0.05}, which is marginally higher than the hardness value of wrought AISI 434L steel. The analysis also shows development of compressive residual stress in the fabricated structure predominantly caused by the complex heating and cooling cycle experienced during the TIG arc scanning. The present study established the potential of TIG PBF-AAM method in the fabrication of a layered structure using metallic powder as feedstock.

Keywords arc additive manufacturing, ferritic stainless steel, microstructure, powder bed fusion, TIG arc

1. Introduction

Additive manufacturing, an automated layer-by-layer fabrication method, is used to create parts from a variety of materials, including polymers, metals, ceramics, and composites. Among these, metal-based AM has arguably had the greatest influence across various sectors, such as medical, aviation, and construction (Ref 1). In metal-based AM, a metallic feedstock material, in the form of wire or powder, is directed toward the heat sources, like laser beam, electron beam, or welding arc (Ref 2), and after melting and subsequent solidification, deposited as a uniform layer. By repeating the process, through layer-by-layer deposition, a 3-dimensional object with desired shape and dimensions is obtained. A metallic additive manufactured part possesses decent mechanical properties and can be used directly for end applications (Ref 3).

In this perspective, laser direct energy deposition (DED) and laser powder bed fusion (PBF) methods are extensively used to fabricate high-precision metallic components (Ref 4, 5), and the fabricated parts exhibited superior mechanical properties as

compared to the wrought alloy (Ref 6). Nevertheless, limited part size, low deposition rate, larger building time, and complexity in the process control are the major downsides of these techniques, which restrain their application to fabricate larger size components (Ref 7).

Recently, arc-based additive manufacturing (AAM) techniques have expanded popularity because of their elevated deposition rate, larger building area, and applicability for a wide range of materials (Ref 8). The typical deposition rate in wire-AAM, working with metal inert gas (MIG) (1–10 kg/h) (Ref 9), tungsten inert gas (TIG) (1–2 kg/h) (Ref 10), and plasma (1.5 kg/h) (Ref 11)-based arc as heat sources is substantially higher than the selective laser melting (0.1 kg/h) and laser metal deposition (1 kg/h) techniques. Henceforth, the wire-AAM technique requires lesser building time and provides a competitive edge over the laser-based AM processes (Ref 12). Also, the tensile properties of various graded steel parts fabricated by wire-AAM technique are superior than that of wrought alloy or parts fabricated by laser-based AM methods (Ref 13). In addition, the wire-AAM technique exhibited features like, high material utilization ratio, eco-friendly production, and low equipment cost (Ref 14). Irrespective of these benefits, distortion in the manufactured part, requirement of specific graded filler wires, and incompetence to fabricate functionally graded component, confine the application of wire-AAM technique for precision part fabrication with a variety of materials. In MIG-based wire-AAM, high spatter formation augmented the surface roughness and waviness of the fabricated part (Ref 15). Further, requirement of a dynamic and hot wire-feeding arrangement and synchronization of the wire feeding with the arc motion are the foremost challenges in TIG or plasma-based wire-AAM methods (Ref 16, 17).

M.D. Aseef Khan and Manoj Masanta, Department of Mechanical Engineering, National Institute of Technology Rourkela, Rourkela 769 008, India. Contact e-mail: manoj.masanta@gmail.com.

Recently, the feasibility of powder-blown AAM was explored with plasma arc, where stainless steel (Ref 18), high entropy alloy (Ref 19), and cobalt-based alloy (Ref 20) powders were employed as feedstock materials. These studies demonstrated the ability of the process to fabricate parts with high deposition efficiency and minimum sidewall roughness. However, non-uniform deposition, inefficient supply of feedstock powder, deflection of plasma arc due to accretion of spatter near the nozzle hole, and porosity within the fabricated parts are the major concerns noticed in the process. To minimize these hitches of powder-blown method, Moreira et al. (Ref 21) established the concept of TIG-based powder bed fusion-type arc additive manufacturing (TIG PBF-AAM) technique, where the metallic powder or powder mixture of a specific thickness was preplaced on a substrate, and TIG arc was scanned over the powder bed, which, after melting and solidification, deposited as a thick layer. With this technique, they fabricated thin wall of H13 tool steel by depositing ten numbers of layers. The individual layer thickness was measured approximately 1.5 mm, which is much higher than that typically obtained in laser-based PBF-AM (200 μm) (Ref 22). Dong et al. (Ref 23) also fabricated a high entropy alloy (HEA) thin wall of dimensions $125 \times 8 \times 25 \text{ mm}^3$ by depositing a total 15 numbers of 1-mm-thick individual layers using TIG PBF-AAM method. In a separate work, to attain homogenous composition and structure of high entropy alloy, interlayer re-melting was performed prior to the deposition of individual layers (Ref 24). The TIG PBF-AAM method is found to be highly suitable for layer-by-layer deposition, as the method is working with similar mechanism of powder bed selective laser sintering or melting method. Also, this technology can accommodate a wide range of metal-based powders, including various alloys and reinforcements, and mixture of metal and ceramic powder, enabling the method suitable for manufacturing custom alloys/composites. Till date very limited literature is available where TIG PBF-AAM technique is used. Considering the prospects of this technique, it can be an alternative to expensive laser-based AM, where large size part with lesser dimensional accuracy can be accepted. The prior studies demonstrated that the method could deposit powder layers of 1-1.5 mm thickness in single pass (Ref 21, 23), highlighting its ability to form 3-dimensional parts with a high building rate.

AISI 434L ferritic stainless steel is an Fe-Cr-based alloy, possesses high tensile strength (538 MPa), excellent oxidation resistance at elevated temperature, formability, and superior machinability but marginally lower corrosion resistance as compared to the austenite stainless steel (Ref 25). Also, higher thermal conductivity and lower thermal expansion coefficient, make it more resistant to thermal fatigue and oxide spalling. Accordingly, the material is used to fabricate components experiencing cyclic heating and cooling in open atmosphere (Ref 26), i.e., automobile exhaust system, combustion chamber, furnace part, heat exchanger, and chemical processing equipment (Ref 27). Numerous studies have demonstrated the effectiveness of different types of AM techniques to fabricate various graded steel components. For instance, Larimian et al. (Ref 28) utilized the selective laser melting technique to fabricate an AISI 316L SS part, where they found laser power and scan speed are the most crucial factors influencing the microstructure and mechanical behavior of the produced part. Hengsbach et al. (Ref 29) also fabricated duplex stainless steel (DSS) part by selective laser melting process, which exhibited

enhanced ultimate tensile strength (UTS) as compared to the wrought alloy.

Nevertheless, very limited studies have explored the AM technique to produce ferritic stainless-steel components. Recently, Karlsson et al. (Ref 30) have yielded ferritic stainless steel parts by laser-assisted PBF-AM technique and witnessed superior tensile strength and toughness as compared to the cast/rolled steel. Meng et al. (Ref 31) observed some anisotropy in the microstructure of high-strength 430 ferritic stainless steel part fabricated by selective laser melting technique, although they achieved adequate tensile properties.

The literature revealed that so far very limited researchers have explored the viability of powder bed fusion-type additive manufacturing using TIG arc as heat source and no specific work reported the use of AISI 434L stainless steel powder in arc-based AM technique. The previous works mainly demonstrated the potential of the TIG PBF-AAM method for fabricating thin wall structures of H13 tool steel and high entropy alloys. The detail microstructural analysis of the fabricated parts in the building direction, which is significant to comprehend the deposition characteristics in different layers, and to improve the fabrication process, is rarely discussed in the available literature. Considering the limitations of wire-AAM and laser-based AM techniques, necessity of an efficient alternative arc-based additive method becomes inevitable.

In this work, an attempt has been made to fabricate a 3-dimensional structure with AISI 434-L stainless steel by using TIG-based powder bed fusion-type arc additive manufacturing (TIG PBF-AAM) method, and study their macro- and microstructural characteristics, along with specific mechanical properties for the variation of processing conditions.

2. Experimental Procedure

A Fronius Magicwave 2200 TIG welding machine has been modified to develop a semi-automated TIG PBF-AAM setup as illustrated in Fig. 1. The TIG torch is attached with a moving tractor enabling automated X-direction motion. The work-table has given a provision to move in Z-direction with the aid of a screw-jack, which is again mounted on a workbench and provides Y-direction motion to the work table.

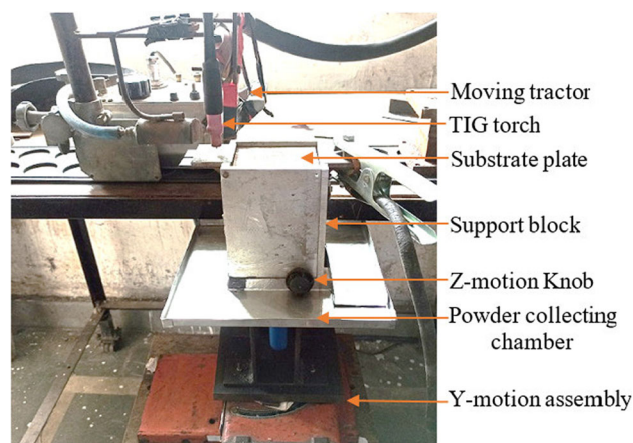


Fig. 1 Experimental setup for TIG-based powder bed fusion-type arc additive manufacturing (TIG PBF-AAM) method

AISI 434L stainless steel powder (particle size: 200–250 μm) was used as feedstock material. The SEM image and EDS analysis of the powder are represented in Fig. 2, whereas its chemical composition, as provided by the manufacturer and obtained through EDS analysis, is depicted in Table 1.

AISI 304 stainless steel plate of 4 mm thickness was taken as a substrate to deposit the layers through TIG PBF-AAM method. Powder preplacement was done by spraying a specific amount of powder on the substrate plate and sweeping the excess powder with the aid of a glass slider. The desired powder layer thickness was maintained by lowering the substrate plate in z-direction against the top surface of the support block. After applying a pre-defined thickness of powder layer, the TIG torch was scanned over the powder bed to attain a single-line deposition. Figure 3 demonstrates the pre-placed powder bed before the TIG arc scanning and the deposited layer after the TIG arc scanning. To identify a suitable processing condition, during TIG arc scanning, the welding current and scan speed were varied by keeping the distance between electrode and powder bed, and gas flow rate constant at 3 mm and 10 L/min, respectively. Accordingly, trial runs were made to deposit single-line tracks considering the parameter combinations as presented in Table 2.

To attain multi-layer deposition, after the formation of 1st layer, the substrate plate was lowered in accordance with the required powder layer thickness, and a new powder layer was applied over the preceding solid layer. To fabricate a 3-dimensional part, the steps were repeated, and five consecutive layers were deposited according to the processing conditions detailed in Table 3, which were selected based on the literature available and analysis of single-layer depositions.

After the single- and multi-layer depositions through arc scanning in TIG PBF-AAM method, the residual powder was removed, and loosely bonded powder particles adhered to the deposited structure and substrate were cleaned. To carry out various metallurgical characterization, the deposited single-line tracks and five-layer structures were cut in cross-sectional direction with the aid of a wire-EDM, and followed by

mounting the specimens in resin mold, and polishing through various graded emery papers, alumina, and diamond slurry sequentially. The polished surfaces were etched in a solution made of hydrochloric acid, hydrogen peroxide, and distilled water (0.3: 0.15: 1 ratio) for 12 seconds. The SEM images of the polished surfaces were captured in backscattered electron imaging (BEI) mode using a JEOL JSM-6480 LV scanning electron microscope. The XRD characteristic peaks of the polished surfaces of the multi-layer deposited specimens were obtained through Bruker AXS D6 advance diffractometer, to determine the phase composition and their quantitative analysis. Co target was used to generate the x-ray, and scanning was executed with a step size of 0.02 and scan rate of 10°/min within the diffraction angle of 30° to 120°. Additionally, the residual stress developed in the fabricated five-layer part, was assessed through XRD technique by $\sin^2\psi$ method. The Vicker's microhardness tester (LECO LM-810) was used to measure the microhardness value of the fabricated structure in the building direction, for which 50-g load and 10 second dwell time were considered.

3. Results and Discussion

3.1 Single-Layer Deposition

Figure 4 shows the images of single-layer depositions yielded with AISI 434L stainless steel powder on AISI 304 steel plate for using different welding current and scan speed combinations (as detailed in Table 2) and the cross-sectional SEM images of the depositions for selected tracks. Because of discontinuity and lack of uniformity, tracks T1 and T2 (deposited at lower current settings) have not been considered for further analysis. From the cross-sectional SEM images, the deposited layer width, height, dilution depth, and contact angle were measured, and the data are plotted against different processing conditions, as illustrated in Fig. 5. The plots indicate that for a constant welding current, increasing the scan speed resulted in a reduction in the weld bead width, dilution depth

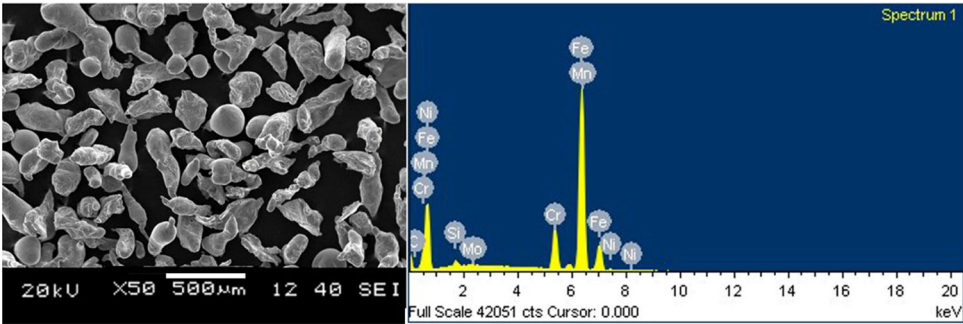


Fig. 2 SEM image and EDS spectra of AISI 434L stainless steel powder

Table 1 Chemical composition of AISI 434L stainless steel powder

Element, %	C	Mn	P	Si	Ni	Cr	Mo	S	Nb	Fe
Supplier's catalog	0.02	0.76	0.17	2.41	0.55	13.2	0.25	0.03	...	Balance
EDS analysis	0.33	0.97	0.16	1.19	0.77	12.93	...	0.12	0.17	Balance

(except for 80A current), and contact angle, but enhancement in the weld bead height. This behavior in the bead geometry attributed to reduced interaction time of the TIG arc with the powder layer. However, for 80A current setting, the weld bead width was found marginally higher for increasing the scan speed from 4.5 mm/s to 5.5 mm/s. Although it is expected that with the increase in scan speed the weld bead width should reduce, in the present case, due to some other factors like, powder spattering during TIG arc scanning and argon gas flow, some localized discrepancy occurred, and some randomness in the weld bead width and dilution depth was observed. The main objective of this set of experiment is to identify the acceptable range of parameter condition considering the height, width, dilution depth, and contact angle of the weld bead, so that in the subsequent section, multilayer deposition could be done with this parameter combination.

To attain high accuracy in the AM fabricated part, controls over layer geometry are highly anticipated. As suggested by Benakis et al. (Ref 32), for wire arc additive manufacturing

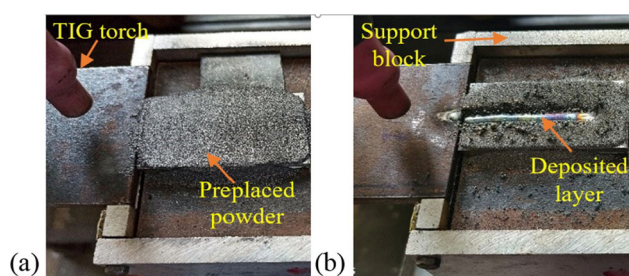


Fig. 3 (a) Pre-placed powder layer before TIG arc scanning, (b) deposited layer after TIG arc scanning

(WAAM), the ideal layer deposition should maintain larger height to width ratio, sufficient dilution depth, minimum heat-affected zone, and least residual stress in the deposited layer. Accordingly, a minimum possible heat input during the arc scanning is recommended. The cross-sectional images of the single-layer depositions (Fig. 4b) and corresponding data (Fig. 5) revealed that low to medium welding current and higher scan speed combinations engender the least possible heat input. Consequently, layer with larger height to width ratio, sufficient dilution depth, and lower contact angle were formed. Henceforth, welding current and speed combinations of 80A-5.5 mm/s (T8) and 100A-5.5 mm/s (T9) were initially chosen for multi-layer deposition.

However, with these combinations, significant discontinuity in the deposition was witnessed, especially at the top layers. An in-depth study on the deposition mechanism in AM with various materials revealed that the discontinuity in the upper layer predominantly occurred because of inadequate width of the preceding layer. Accordingly, a wider base layer was expected to be more suitable for the continuous deposition of successive layers. Henceforth, in the subsequent experiments, the depositions were executed by increasing the welding current and scan speed simultaneously, so that almost uniform heat input is to be maintained. Figure 6 illustrates the images of five-layer deposited structures for different processing conditions, as detailed in Table 3.

Figure 6 shows that for specimens H1 and H2 (fabricated at 80A-5.5 mm/s and 100A-5.5 mm/s welding current and scan speed combinations), the final depositions are intermittent type, especially at the upper layers. For specimen H3 (deposited at 100A-3.5 mm/s), the discontinuity reduced significantly with minor variation in the deposited height. For specimens H4, H5, and H6 (processed at relatively higher welding current and scan

Table 2 Processing conditions used for single-layer depositions

Track no.	Processing conditions			
	Welding current, A	Voltage, V	Scan speed, mm/s	Heat input, kJ/mm
T1	60	18.6	2.5	0.290
T2	60	18.5	3.5	0.206
T3	80	19.5	3.5	0.290
T4	100	20.2	3.5	0.375
T5	120	20.5	3.5	0.457
T6	140	21.1	3.5	0.549
T7	80	19.3	4.5	0.223
T8	80	19.5	5.5	0.184
T9	100	19	5.5	0.225

Table 3 Processing conditions for five-layer deposition

Specimen	Processing conditions			
	Welding current, A	Corresponding voltage, V	Scan speed, mm/s	Heat input, kJ/mm
H1	80	19.6	5.5	0.185
H2	100	19.9	5.5	0.235
H3	100	16.3	3.5	0.303
H4	120	17.4	4.5	0.302
H5	140	18.9	5.5	0.313
H6	160	20.1	6.5	0.322

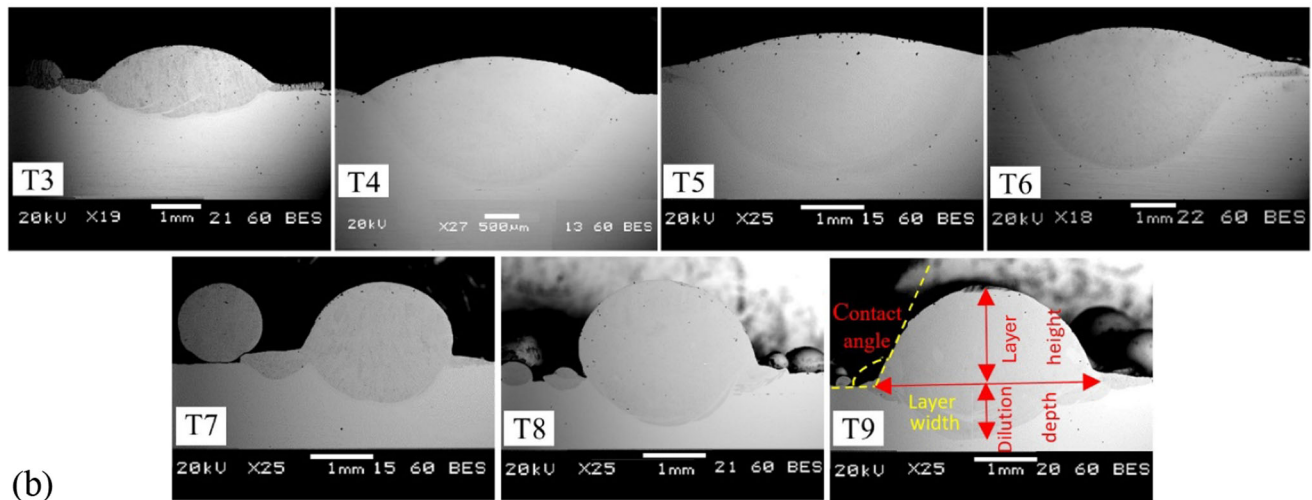
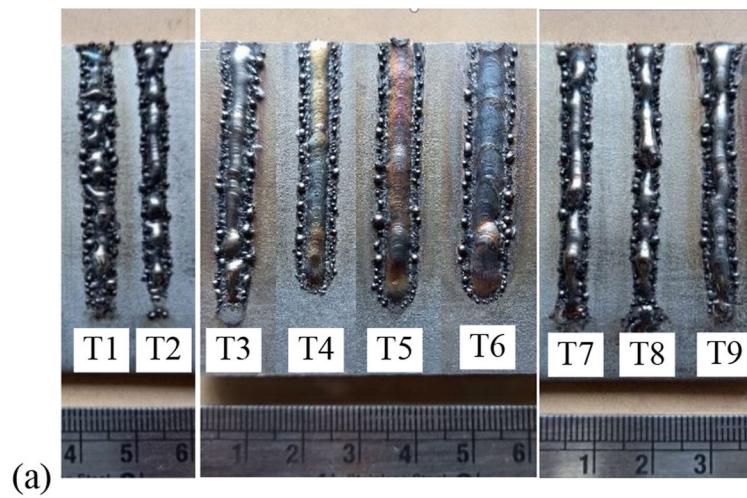


Fig. 4 Single-layer depositions of AISI 434L stainless steel powder by TIG PBF-AAM method for using different welding current and scan speed as detailed in Table 2: (a) optical image from top surface and (b) cross-sectional SEM images

speed combinations), almost smooth and uniform depositions were obtained. Henceforward, specimens H4, H5, and H6 were considered for further analysis of structural geometry, phase composition, microstructure, microhardness, and residual stress. The investigation indicated that for increasing the welding current, the uniformity and steadiness in the deposited layers augmented expressively, and for specimen H6 (160A-6.5 mm/s), almost uniform depositions with minimum deviation in the height and thickness of the structure were obtained.

3.2 Cross Section of Multi-Layer Deposition

Figure 7(a-c) illustrates the cross-sectional images from the transverse direction, geometrical assessment (total height, average width, and skewness), and cumulative layer heights of the five-layered structures (specimen H4, H5, and H6) fabricated by TIG PBF-AAM method through multilayer deposition. The cross-sectional images indicating reasonably narrower width of the initial or bottom layers than those of the upper layers. As the substrate plate is acting as a heat sink, during solidification of the bottom layers, the cooling rate expedited significantly and solidified the melt-pool instantly, which in turn restrain the extent of molten material in lateral direction and confine the deposited layer width. In contrast,

during the deposition of upper layers, the cooling rate is judiciously slower, as the heat is accumulated due to repetition of the arc scanning. Thus, the resulted molten pool is allowed to flow in the lateral direction and enhanced the deposited layer width. Also Fig. 7(b) indicates that for increasing both the welding current and scan speed simultaneously, the average width of the deposited structure reduced almost linearly, whereas a marginal augmentation in the total height of the deposited structure is witnessed.

Further, a marginal reduction in the total height of the fabricated part is observed, when compared with the total deposited powder layer thickness. It is evident that after the deposition of five layers, approximately 7.4, 3.87, and 3.6% reduction in the total layer height as compared to the total deposited powder layer thickness, occurred in specimens H4, H5, and H6, respectively. Upon exposure of the powder bed with the TIG arc, owing to the arc pressure and shielding gas flow, there is a possibility of spattering of the preplaced powder, which causes deficiency in the powder to be deposited. Additionally, the powder layer undergone through melting and solidification cycle, which eliminates the inter-particle gap or voids after solidification, that resulted in compactness or shrinkage in the re-solidified layer. The powder layer also

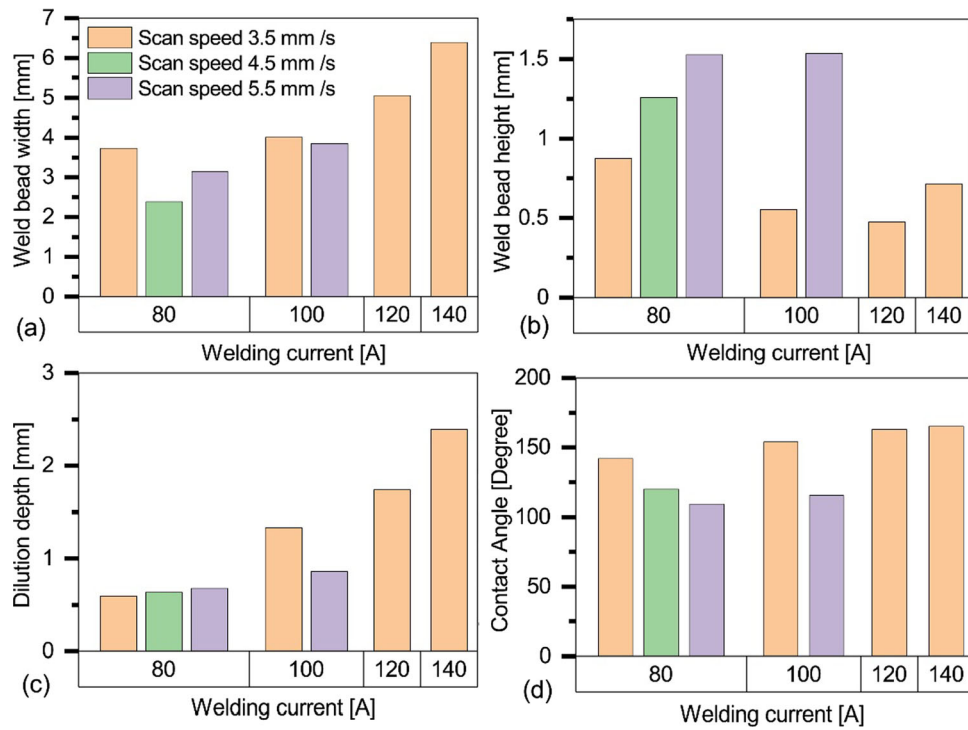


Fig. 5 Variation of (a) layer width, (b) layer height, (c) dilution depth, and (d) contact angle for single-layer depositions of AISI 434L stainless steel obtained through TIG PBF-AAM method for varying welding current and scan speed combinations

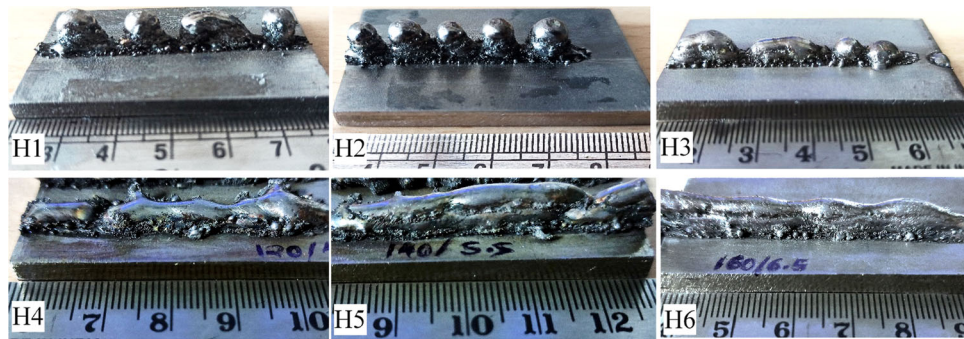


Fig. 6 Five-layer deposited AISI 434L structure obtained through TIG PBF-AAM method for different processing conditions as detailed in Table 3

experienced phase transformation during the solid-liquid-solid conversion due to intricate heat cycles involved in the TIG PBF-AAM method, which also leads to marginal shrinkage in the deposited material.

The skewness, which is the measure of asymmetry, was measured from the cross-sectional images of the deposited layer. The plotted values (Fig. 7b) indicated a larger asymmetry in the deposited structure for higher processing current. It is expected that the melt pool of the pre-deposited powder flows in a lateral direction due to high arc pressure. During subsequent deposition, some deviation in the alignment of the powder layer also occurred. Further, owing to inadequate powder compaction and instability in the TIG arc, a non-uniform deposition is possible. The lateral movement of the molten metal, the scanning track deviation due to alignment of the TIG torch, and non-uniform deposition due to variation in the preplaced powder layers, collectively induce skewness in the fabricated part. At higher processing current, due to

superheating of the molten pool, its viscosity is expected to reduce, and the arc pressure increases, which possibly augmented the lateral flow of the molten metal, and amplified the skewness in fabricated part.

The plot in Fig. 7(c) indicated that the initial or bottom layer thicknesses are relatively smaller as compared to the upper layers. The molten powder undergoes through a volumetric reduction during fusion of powder bed through TIG arc scanning and subsequent solidification due to various reasons as discussed earlier. Thus, a relatively lower deposition height as compared to the applied powder layer thickness was obtained. However, during powder preplacement, the layer thickness was controlled by lowering the substrate against the previous position. Accordingly, it is obvious that the applied powder layer thickness enhanced for successive depositions, during the fabrication of a multi-layer part. Consequently, the thickness or height of the subsequent upper layers was found reasonably larger than the lower layers. This phenomenon can

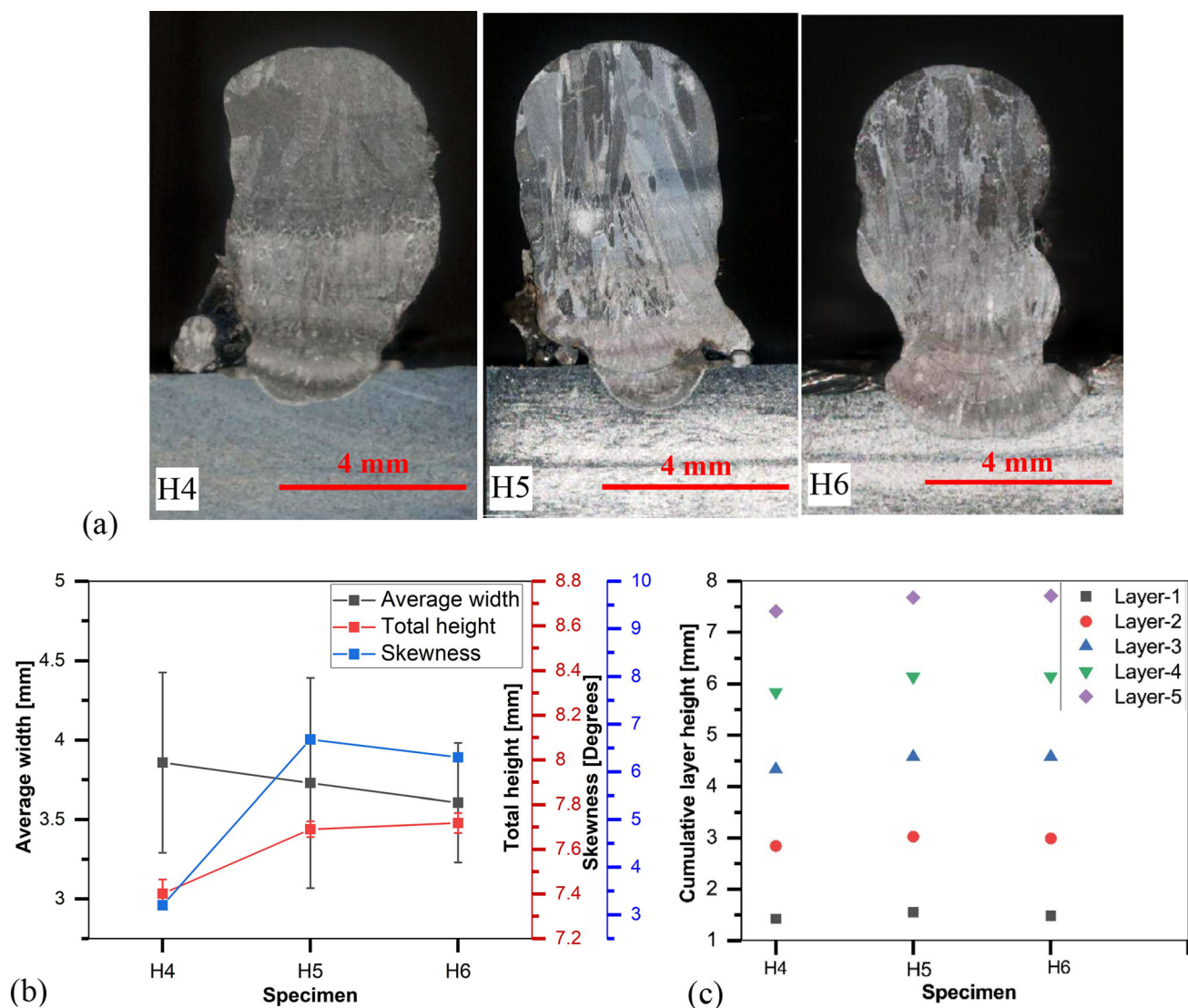


Fig. 7 (a) Cross-sectional images, (b) geometrical assessment (average width, total height, and skewness), and (c) cumulative layer height of the five-layer deposited AISI 434L stainless steel structures fabricated by TIG PBF-AAM method

Table 4 Calculation of deposition rate in TIG PBF-AAM method

Sample	Deposited weight, g	Deposition time, s	Deposition rate, kg/hr
H4	12.85	65	0.711
H5	12.29	53	0.835
H6	13.37	45	1.07

be controlled by appropriate movement of the base plate, considering the reduction of individual layer height.

The deposition rate in additive manufacturing is a vital factor, as it reflects the efficiency, building rate, and productivity of the process. In this aspect, the deposition rate for the specimens (H4, H5, and H6) processed at different current and scan speed combinations was assessed by dividing the total deposited weight with the deposition time. The details of the calculation illustrated in Table 4, show that for using AISI 434L stainless steel powder, and within the experimental domain, the magnitude of deposition rate is in the range of 0.71 – 1.07 kg/

hr. These values are comparable to the deposition rate of TIG-based wire-AAM (1-2 kg/hr) (Ref 11). The data also indicate that the deposition rate increases almost gradually with the welding current. A higher welding current creates a larger melt pool, allowing a larger volume of powder to be melt, and exhibited a higher deposition rate.

3.3 Microstructure of the Deposited Structure

The SEM images of the top, middle, and bottom regions captured from the transverse cross section of the fabricated

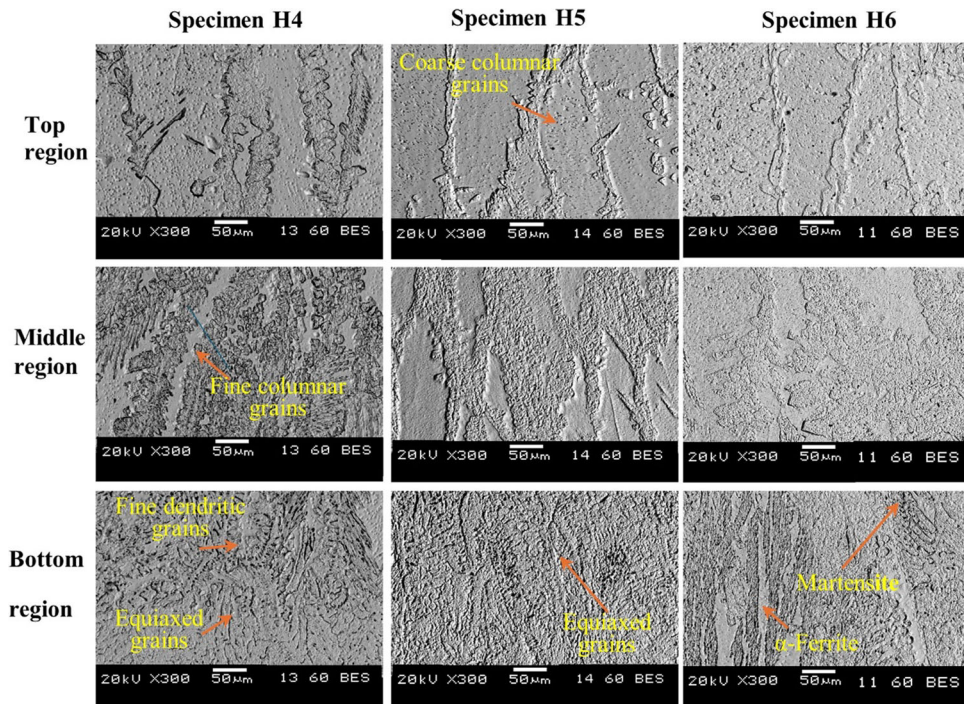


Fig. 8 Microstructure of the top, middle, and bottom region of five-layer AISI 434L SS structure fabricated by TIG PBF-AAM method

structures (specimen H4, H5, and H6) are illustrated in Fig. 8. The images show that fine dendrites and randomly oriented equi-axed grains are present in the bottom region, where the dendrites are predominantly aligned along the building direction. However, the middle and top regions are comprised of fine and coarse columnar grains, respectively. The transformation of fine to coarse columnar grain-structure from the middle to top region of the deposited layers occurred due to larger accumulation of heat in the succeeding layers and corresponding slower cooling of the deposited material during solidification. Again, the successive deposition resulted in re-melting or reheating of the preceding layer, which enable to form columnar grains parallel to the building direction. However, at the top region the grains for the individual layers are not distinguishable, and a continuous structure appeared along the building direction. Laghi et al. (Ref 33) also perceived formation of coarse columnar grains at the top regions of thin wall fabricated with austenitic stainless steel through wire arc additive manufacturing (WAAM) method. Karlsson et al. (Ref 34) and Durga et al. (Ref 35) also witnessed long columnar grains aligned along the building direction in ferritic stainless steel parts fabricated by laser PBF-AM method. They explained the phenomenon as grain growth along the direction of thermal gradient due to temperature variation in the successive layers. The AISI 316L SS block fabricated by selective laser melting technique also exhibited coarse dendrites oriented in the building direction. The phenomenon was described as the effect of elevated laser power and sluggish cooling rate of the laser processing (Ref 36). In another study, Meng et al. (Ref 31) observed that 430 ferritic stainless steel specimens fabricated through selective laser melting process, exhibited microstructural inhomogeneity (fine, coarse, equiaxed, and columnar grains), attributed to occurrence of disparate heat transfer within various regions of the specimen.

Table 5 Phase content (%) of AISI 434L raw powder and fabricated specimens obtained through relative intensity method (RIM) and Rietveld refinement method (RRM)

Specimen	Ferrite, %		Martensite, %	
	RIM	RRM	RIM	RRM
AISI 434L powder	83.56	83	16.44	17
H4 (Bottom)	46.36	40.3	53.63	59.7
H5 (Bottom)	49.4	44.2	50.58	55.8
H6 (Bottom)	53.22	48.9	46.77	51.1
H6 (Top)	93.66	85	6.33	15

The analysis of the microstructure also revealed that for all the specimens, the deposited layers are predominantly consisted of α -ferrite (gray-shaded regions) along with martensite phase (black color regions) in the grain boundaries. The martensite phase transformation in the solidified structure occurred due to steep thermal gradient and rapid cooling caused by the heat conduction through the substrate material. Accordingly, at the bottom region of the fabricated structure, the martensite phase content is found to be more significant as compared to the top region, which is supported by the XRD analysis data discussed in the subsequent section (Table 5). Ramkumar et al. (Ref 37) also witnessed α -ferrite and martensite phases within the fusion zone during TIG welding of AISI 430 ferritic stainless steel. Jiang et al. (Ref 38) and Karlsson et al. (Ref 30) also observed nearly similar microstructure in ferritic stainless steel components fabricated by selective laser melting and laser powder bed fusion methods separately.

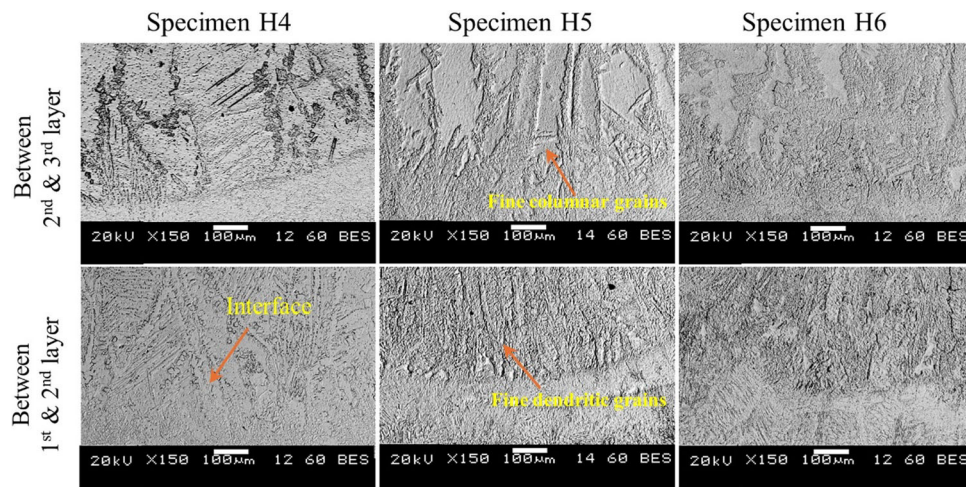


Fig. 9 Microstructure at the interface between 1st-2nd layer and 2nd-3rd for five-layer deposited specimens

The SEM images taken at the interfaces between the successive layers, as illustrated in Fig. 9, revealed the emergence of fine dendritic grains between the 1st and 2nd layers, whereas fine columnar grains can be seen at the interface between 2nd and 3rd layers. Also, no interfacial defects such as lack of fusion, delamination, or inclusion of external material are detected, which indicates the formation of a robust bond between the successive layers.

3.4 XRD Analysis

The compositional changes occurred during the fabrication of AISI 434L steel structure by TIG PBF-AAM method, were analyzed through XRD technique executed on as-received AISI 434L powder and the fabricated structure. The XRD patterns for AISI 434L powder and the fabricated specimens (H4 to H6) are illustrated in Fig. 10. The characteristic peak at 2θ positions of 52° and 99° assigned to [110] and [211] planes, indicates the presence of α -ferrite (JCPDS No. 06-0696) and martensite (JCPDS No. 44-1292) phases, while the peak at 77° corresponds to [200] plane and represents the α -ferrite phase. Following the phase identification, the phase fractions are also quantitatively analyzed using the Rietveld refinement method and the relative intensity method, and results are summarized in Table 5.

The data indicate that the phase fraction of α -ferrite and martensite, analyzed through relative intensity method and Rietveld refinement method, is almost identical. Also, the martensite content at the bottom region of the fabricated parts increased nearly three times after the deposition of AISI 434L SS powder through TIG PBF-AAM method. However, the martensite percentage at the bottom region of the fabricated structure reduced meaningfully from specimen H4 to H6, where welding current enhanced from 120 A to 160 A, along with simultaneous increase of the scan speed from 4.5 mm/s to 6.5 mm/s. At higher welding current, because of high heat input, during solidification of the deposited material, the cooling rate was also reduced, and possibility of martensite phase formation became moderated.

A significant variation in the phase content was also observed at the bottom and top region of the structure as analyzed for the specimen H6. The α -ferrite content near the top layers was recorded substantially higher, which is almost

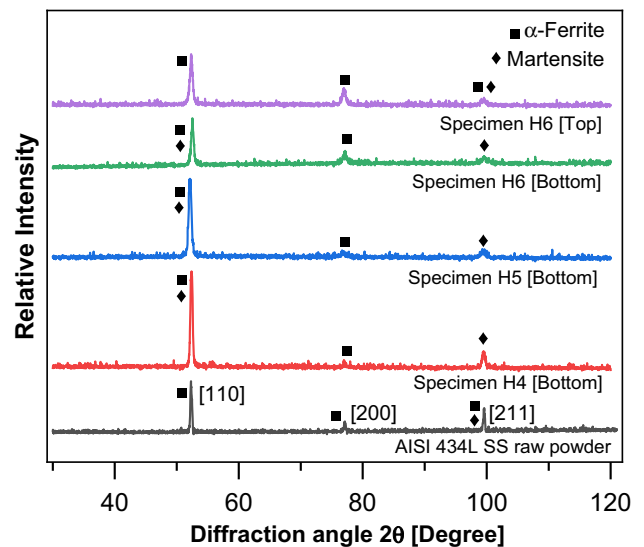


Fig. 10 XRD patterns of as-received AISI 434L stainless steel powder and specimens fabricated by TIG PBF-AAM method

comparable with the as-received AISI 434L SS powder. As the number of deposited layers increased, the fabricated structure was experiencing more number of heating cycles, which resulted in heat accumulation, and consequently depressed the cooling rate, and augmented the possibility of α -ferrite phase formation.

3.5 Microhardness Analysis

The microhardness was measured in line from the top to bottom of the fabricated structure at an interval of $450\ \mu\text{m}$. For each traverse position, the average microhardness value was calculated considering three readings and is plotted in Fig. 11. The plot indicates that the average microhardness values are within a range of 186-330 $\text{HV}_{0.05}$, with some inconsistency along the building direction. This variation is mainly attributed to non-homogenous microstructure and disparity in the phase composition in the fabricated structure toward the building direction. The plot also revealed that the microhardness values

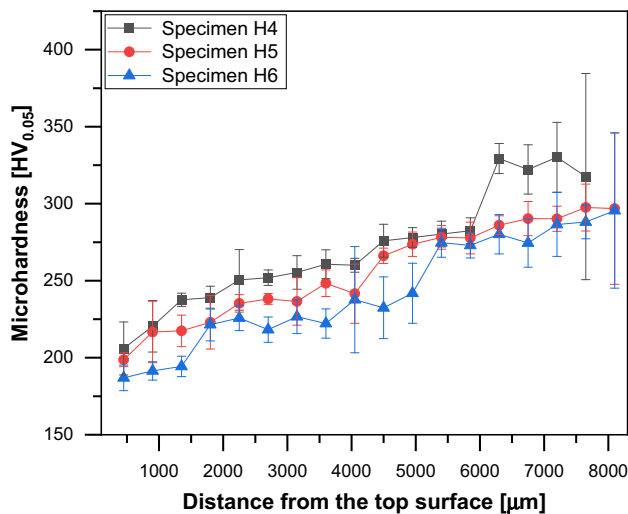


Fig. 11 Variation of microhardness measured from the top surface for the TIG PBF-AAM fabricated specimens

at the bottom regions are reasonably higher as compared to the top region of the fabricated parts.

The microhardness value in a component is largely governed by its microstructure or the presence of any irregularity within it. As revealed through the microstructural analysis, the top region of the fabricated part consisted of coarse columnar grains with limited grain boundaries, leading to lower microhardness values. Conversely, the fine dendritic or equi-axed grains at the bottom region resist the dislocation motion during the indentation and exhibited reasonably higher microhardness value. The plot also demonstrates a progressive reduction in the microhardness value toward the building direction.

The phase composition of the deposited material is the most influential factor for the variation of microhardness values in the fabricated part. A higher fraction of martensite phase at the bottom region is also responsible for higher microhardness as compared to the top region. Xie et al. (Ref 39) also revealed that in AISI 316L steel part, fabricated by cold metal transfer method, the hardness values near the bottom region are much higher than the top region. Almost similar variation in the microhardness value was also reported by Venkatesh et al. (Ref 40) in 409L ferritic stainless steel weld joint executed by P-TIG welding method. They attributed this discrepancy as an outcome of martensite transformation at the bottom region during the welding process. It is evident that for specimen H6, the overall microhardness values are sensibly lower than that of specimen H4. This is due to yielding of larger size grains under higher welding current as detailed in microstructural analysis. Also, a reduction in the martensite phase fraction resulted in a reduced microhardness in specimen H6.

The average microhardness values for the fabricated parts by TIG PBF-AAM method are almost equivalent to TIG welded 430 ferritic stainless steel joints (250.6 HV_{0.05}) (Ref 41). In the present work, through TIG PBF-AAM technique, the microhardness of the fabricated AISI 434L stainless steel specimen was obtained upto 330 HV_{0.05}, which is notably lesser than the laser-based AM fabricated ferritic stainless steel component (812.8 HV_{0.2}) reported by Jiang et al. (Ref 38) and expected to be beneficial by restraining the brittleness of the final component.

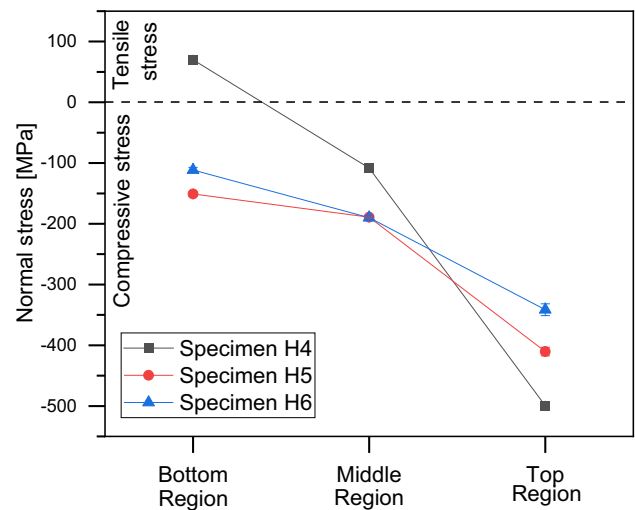


Fig. 12 Normal residual stress at various locations in the building direction for the TIG PBF-AAM fabricated parts (specimen H4, H5, and H6) as measured through XRD technique

3.6 Residual Stress

The major concern in the metal-based additive manufactured part is development of residual stress, because of complex thermal cycle of the process. The performance of a component degrades due to distortion and micro-crack formation when the residual stress exceeds a certain value. Hence, to ensure the quality of the part, evaluation of residual stress in a AM part is greatly significant.

The residual stress of the fabricated five-layer parts is assessed through XRD technique using $\sin^2\psi$ method. From the XRD pattern, [211] lattice plane was selected, and the modulus of elasticity and Poisson's ratio for AISI 434L stainless steel were considered as 200 GPa and 0.28, respectively. The $\sin^2\psi$ vs lattice spacing (d) plots were obtained by XRD technique, and the normal residual stress values were assessed at different locations of the specimens along the building direction, as illustrated in Fig. 12.

It is revealed that the normal residual stress at the bottom regions of the fabricated parts is very close to zero. For specimen H4, it is tensile (+70 MPa), while for specimens H5 and H6, the values are compressive (−150 and −111 MPa). However, toward the building direction, the developed residual stresses are recorded as compressive for all the specimens. At the middle region of the fabricated part, the measured residual stresses are in the range of 108–190 MPa, while at the topmost region, the values are within a range of 341–499 MPa.

During TIG arc scanning, the melt pool of the AISI 434L stainless steel powder contracted, as it solidifies after the initial layer was applied on the substrate, which leads to a compressive stress within the deposited layer. The tensile stress is simultaneously generated close to the substrate due to variations in the thermal expansion rate. These high-temperature stresses steadily decline as the deposited layer becomes cool down and minimizes both the tensile and compressive stresses. In consequence, at the bottom region, a relatively lower compressive stress (specimens H5 and H6) or minor tensile stress is developed (specimen H4). As the number of layers increased, the preceding deposited layers are undergone through repetitive heating and cooling cycle, and cause a

significant temperature variance between the bottom and top regions that leads to a steep thermal gradient in the building direction. Consequently, the cooling rate of the molten pool also varies, which causes differential contraction of the deposited layers in the building direction, and augmented the compressive residual stress. Mishurova et al. (Ref 42) in their work also recorded tensile residual stress near the bottom region, almost zero residual stress at the middle zone, and high compressive residual stress close to the top region for a Ti6Al4V wall fabricated by wire-AAM technique. Kumaran et al. (Ref 43) and Gordon et al. (Ref 44) also observed similar behavior in the residual stress measured in building direction for the austenitic stainless steel parts fabricated by laser direct energy deposition and wire-AAM process.

4. Conclusions

The study revealed the feasibility and potential of TIG-based powder bed fusion-type arc additive manufacturing (TIG PBF-AAM) technique to fabricate a thin wall structure using AISI 434L ferritic stainless steel powder.

With suitable TIG welding parameter combinations, the method enables to deposit almost uniform layers. No specific interlayer defects, such as cracks, porosity, and delamination, were found in the deposited part, which represents a robust bonding between the consecutive layers.

The deposition rate of the present TIG PBF-AAM method for AISI 434L ferritic stainless steel reached upto 1.07 kg/h, which is comparable to standard TIG wire-AAM method (1–2 kg/h), shows the efficiency of the process.

The deposited layers are consisted of α -ferrite and martensite phases, with some microstructural variation in the building direction. Mostly, fine dendrites and randomly oriented equiaxed grains are formed at the bottom regions, while large columnar grains are observed at the top regions of the fabricated parts.

The microhardness values of the fabricated part are obtained in the range of 186–330 HV_{0.05}, which is marginally higher than the wrought AISI 434L stainless steel, but almost 67% lesser than the ferritic stainless steel part fabricated through selective laser melting by another research group. Also, a moderate variation in the hardness value indicates some anisotropy in the mechanical properties along the building direction of the fabricated part.

A compressive residual stress developed in the fabricated part (108–499 MPa) mainly due to complex heating and cooling cycles and steep thermal gradient along the building direction.

References

1. M. Armstrong, H. Mehrabi, and N. Naveed, An Overview of Modern Metal Additive Manufacturing Technology, *J. Manuf. Process.*, 2022, **84**, p 1001–1029
2. A. Gisario, M. Kazarian, F. Martina, and M. Mehrpouya, Metal Additive Manufacturing in the Commercial Aviation Industry: A review, *J. Manuf. Syst.*, 2019, **53**, p 124–149
3. B. Vayre, F. Vignat, and F. Villeneuve, Metallic Additive Manufacturing: State-Of-The-Art Review and Prospects, *Mech. Ind.*, 2012, **13**(2), p 89–96
4. K. Zhang, S. Wang, W. Liu, and X. Shang, Characterization of Stainless Steel Parts by Laser Metal Deposition Shaping, *Mater. Des.*, 2014, **55**, p 104–119
5. J. Suryawanshi, K.G. Prashanth, and U. Ramamurty, Mechanical Behavior of Selective Laser Melted 316L Stainless Steel, *Mater. Sci. Eng. A*, 2017, **696**, p 113–121
6. W. Huang, Y. Zhang, W. Dai, and R. Long, Mechanical Properties of 304 Austenite Stainless Steel Manufactured by Laser Metal Deposition, *Mater. Sci. Eng. A*, 2019, **758**, p 60–70
7. A. Vranic, N. Bogojevic, S.C. Kostic, D. Crococolo, and G. Olmi, Advantages and Drawbacks of Additive Manufacturing, *IMK-14 Res. Dev. Heavy Mach.*, 2017, **23**(2), p 57–62
8. B. Wu, Z. Pan, D. Ding, D. Cuiuri, H. Li, J. Xu, and J. Norrish, A Review of the Wire Arc Additive Manufacturing of Metals: Properties, Defects and Quality Improvement, *J. Manuf. Process.*, 2018, **35**, p 127–139
9. E. Aldalur, F. Veiga, A. Suarez, J. Bilbao, and A. Lamikiz, High Deposition Wire Arc Additive Manufacturing of Mild Steel: Strategies and Heat Input Effect on Microstructure and Mechanical Properties, *J. Manuf. Process.*, 2020, **58**, p 615–626
10. K.E.K. Vimal, M.N. Srinivas, and S. Rajak, Wire Arc Additive Manufacturing of Aluminium Alloys: A Review, *Mater. Today: Proc.*, 2019, **41**, p 1139–1145
11. I. Tabernero, P. Alvarez, and A. Suarez, Study on Arc Welding Processes for High Deposition Rate Additive Manufacturing, *Procedia CIRP*, 2018, **68**, p 358–362
12. S.W. Williams, F. Martina, A.C. Addison, J. Ding, G. Pardo, and P. Colegrove, Wire + Arc Additive Manufacturing, *Mater. Sci. Technol.*, 2016, **32**(7), p 641–647
13. V.T. Le, D.S. Mai, T.K. Doan, and H. Paris, Wire and Arc Additive Manufacturing of 308L Stainless Steel Components: Optimization of Processing Parameters and Material Properties, *Eng. Sci. Technol. Int. J.*, 2021, **24**(4), p 1015–1026
14. C. Xia, Z. Pan, J. Polden, H. Li, Y. Xu, S. Chen, and Y. Zhang, A Review on Wire Arc Additive Manufacturing: Monitoring, Control and A Framework of Automated System, *J. Manuf. Syst.*, 2020, **57**, p 31–45
15. J. Xiong, Y. Li, R. Li, and Z. Yin, Influences of Process Parameters on Surface Roughness of Multi-Layer Single-Pass Thin-Walled Parts in GMAW-Based Additive Manufacturing, *J. Mater. Process. Technol.*, 2018, **252**, p 128–136
16. E. Spaniol, T. Ungethum, M. Trautmann, K. Andrusch, M. Hertel, and U. Fussel, Development of a Novel TIG Hot-Wire Process for Wire and Arc Additive Manufacturing, *Weld. World*, 2020, **64**, p 1329–1340
17. R.H. Gonçalves e Silva, L.E. dos Santos Paes, G.L. de Sousa, C. Marques, A.B. Viviani, M.B. Schwedersky, and T.L.F. da Costa Pinto, Design of a Wire Measurement System for Dynamic Feeding TIG Welding using Instantaneous Angular Speed, *Int. J. Adv. Manuf. Technol.*, 2019, **101**(5–8), p 1651–1660
18. M. Dalaee, F. Cheaitani, A. Arabi-Hashemi, C. Rohrer, B. Weisse, C. Leinenbach, and K. Wegener, Feasibility Study in Combined Direct Metal Deposition (DMD) and Plasma Transfer Arc Welding (PTA) Additive Manufacturing, *Int. J. Adv. Manuf. Technol.*, 2020, **106**(9–10), p 4375–4389
19. Q. Shen, J. Xue, X. Yu, Z. Zheng, and N. Ou, Powder Plasma Arc Additive Manufacturing of CoCrFeNi High-Entropy Alloys: Microstructure Evolution and Mechanical Properties, *J. Alloys Compd.*, 2022, **922**, p 166245
20. P. Kumar and N.K. Jain, Effect of Material Form on Deposition Characteristics in Micro-Plasma Transferred Arc Additive Manufacturing Process, *CIRP J. Manuf. Sci. Technol.*, 2020, **30**, p 195–205
21. A.F. Moreira, K.S. Ribeiro, F.E. Mariani and R.T. Coelho, An Initial Investigation of Tungsten Inert Gas (TIG) Torch as Heat Source for Additive Manufacturing (AM) Process, *Procedia Manuf.*, 2020, **48**, p 671–677
22. K. Hofer, A. Haelsig, and P. Mayr, Arc-Based Additive Manufacturing of Steel Components—Comparison of Wire- and Powder-Based Variants, *Weld. World*, 2018, **62**, p 243–247
23. B. Dong, Z. Wang, Z. Pan, O. Muransky, C. Shen, M. Reid, B. Wu, X. Chen, and H. Li, On The Development of Pseudo-Eutectic AlCoCrFe-Ni_{2.1} High Entropy Alloy using Powder-Bed Arc Additive Manufacturing (PAAM) Process, *Mater. Sci. Eng. A*, 2021, **802**, p 140639
24. J. Wang, Y. Lu, F. Jia, W. Xia, F. Lin, J. Han, R. Wang, Z. Pan, H. Li, and Z. Jiang, Effects of Inter-Layer Remelting Frequency on the

- Microstructure Evolution and Mechanical Properties of Equimolar CoCrFeNiMn High Entropy Alloys During In-Situ Powder-Bed Arc Additive Manufacturing (PBAAM) Process, *J. Mater. Sci. Technol.*, 2022, **113**, p 90–104
25. J.E. Klar and P.K. Samal, Powder Metallurgy Stainless Steels: Processing, Microstructures, and Properties, *ASM International*, 2007, p 5–19
 26. J.H. Reinshagen and R.P. Mason, Basics of 400 Series P/M Stainless Steels, *Adv. Powder Metall. Part. Mater.*, 1997, **1**, p 9
 27. M. Gurram, K. Adepu, R.R. Pinninti, and M.R. Gankidi, Effect of Copper and Aluminium Addition on Mechanical Properties and Corrosion Behaviour of AISI 430 Ferritic Stainless Steel Gas Tungsten Arc Welds, *J. Mater. Res. Technol.*, 2013, **2**(3), p 238–249
 28. T. Larimian, B. AlMangour, D. Grzesiak, G. Walunj, and T. Borkar, Effect of Laser Spot Size, Scanning Strategy, Scanning Speed, and Laser Power on Microstructure and Mechanical Behavior of 316L Stainless Steel Fabricated via Selective Laser Melting, *J. Mater. Eng. Perform.*, 2022, **31**, p 2205–2224
 29. F. Hengsbach, P. Koppa, K. Duschik, M.J. Holzweissig, M. Burns, J. Nellesen, W. Tillmann, T. Troster, K.P. Hoyer, and M. Schaper, Duplex Stainless Steel Fabricated by Selective Laser Melting-Microstructural and Mechanical Properties, *Mater. Des.*, 2017, **133**, p 136–142
 30. D. Karlsson, C.Y. Chou, N.H. Pettersson, T. Helander, P. Harlin, M. Sahlberg, G. Lindwall, J. Odqvist, and U. Jansson, Additive Manufacturing of the Ferritic Stainless Steel SS441, *Addit. Manuf.*, 2020, **36**, p 101580
 31. L. Meng, W. Li, H. Lu, Q. Zhang, S. Wang, Y. Nie, Y. Wang, W. Liang, and L. Zheng, High-Strength 430 Ferritic Stainless Steel Fabricated by Selective Laser Melting Process, *Mater. Lett.*, 2024, **363**, p 136311
 32. M. Benakis, D. Costanzo, and A. Patran, Current Mode Effects on Weld Bead Geometry and Heat Affected Zone In Pulsed Wire Arc Additive Manufacturing of Ti-6-4 and Inconel 718, *J. Manuf. Process.*, 2020, **60**, p 61–74
 33. V. Laghi, M. Palermo, L. Tonelli, G. Gasparini, L. Ceschini, and T. Trombetti, Tensile Properties and Microstructural Features of 304L Austenitic Stainless Steel Produced by Wire-and-Arc Additive Manufacturing, *Int. J. Adv. Manuf. Technol.*, 2020, **106**, p 3693–3705
 34. D. Karlsson, T. Helander, E. Bettini, C.J. Hassila, J. Cedervall, M. Sahlberg, P. Harlin, and U. Jansson, Relationship Between Microstructure, Mechanical Properties and Creep Behavior of a Cr-Rich Ferritic Stainless Steel Produced by Laser Powder Bed Fusion, *Alloys*, 2022, **1**(3), p 263–276
 35. A. Durga, N.H. Pettersson, S.B.A. Malladi, Z. Chen, S. Guo, L. Nyborg, and G. Lindwall, Grain Refinement in Additively Manufactured Ferritic Stainless Steel by In Situ Inoculation Using Pre-Alloyed Powder, *Scripta Mater.*, 2021, **194**, p 113690
 36. T. Larimian, M. Kannan, D. Grzesiak, B. AlMangour, and T. Borkar, Effect of Energy Density and Scanning Strategy on Densification, Microstructure and Mechanical Properties of 316L Stainless Steel Processed via Selective Laser Melting, *Mater. Sci. Eng. A*, 2020, **770**, p 138455
 37. K.D. Ramkumar, A. Chandrasekhar, A.K. Singh, S. Ahuja, A. Agarwal, N. Arivazhagan, and A.M. Rabel, Comparative Studies on the Weldability, Microstructure and Tensile Properties of Autogeneous TIG Welded AISI 430 Ferritic Stainless Steel With and Without Flux, *J. Manuf. Process.*, 2015, **20**, p 54–69
 38. P.F. Jiang, C.H. Zhang, S. Zhang, J.B. Zhang, J. Chen, and H.T. Chen, Additive Manufacturing of Novel Ferritic Stainless Steel by Selective Laser Melting: Role of Laser Scanning Speed on The Formability, Microstructure and Properties, *Opt. Lasers Eng.*, 2021, **140**, p 107055
 39. B. Xie, J. Xue, and X. Ren, Wire Arc Deposition Additive Manufacturing and Experimental Study of 316L Stainless Steel by CMT+P Process, *Metals*, 2020, 10(11), p 1419
 40. R. Venkatesh, S. Manivannan, S. Kaliappan, S. Socrates, S. Sekar, P.P. Patil, L. Natrayan, and M.B. Bayu, Influence of Different Frequency Pulse on Weld Bead Phase Ratio in Gas Tungsten Arc Welding by Ferritic Stainless Steel AISI-409L, *J. Nanomater.*, 2022, **1**, p 9530499
 41. G. Shanmugasundar, A. Bansod, V. Schindlerova, and R. Cep, Influence of filler Material on the Microstructural and Mechanical Properties of 430 Ferritic Stainless Steel Weld Joints, *Materials*, 2023, **16**(4), p 1590
 42. T. Mishurova, B. Sydow, T. Thiede, I. Sizova, A. Ulbricht, M. Bambach, and G. Bruno, Residual Stress and Microstructure of a Ti-6Al-4V Wire Arc Additive Manufacturing Hybrid Demonstrator, *Metals*, 2020, **10**(6), p 701
 43. M. Kumaran, V. Senthilkumar, C.J. Panicke, and R. Shishir, Investigating the Residual Stress in Additive Manufacturing of Repair Work by Directed Energy Deposition Process on SS316L Hot Rolled Steel Substrate, *Mater. Today: Proc.*, 2021, **47**, p 4475–4478
 44. J.V. Gordon, C.V. Haden, H.F. Nied, R.P. Vinci, and D.G. Harlow, Fatigue Crack Growth Anisotropy, Texture and Residual Stress in Austenitic Steel Made by Wire And Arc Additive Manufacturing, *Mater. Sci. Eng. A*, 2018, **724**, p 431–438

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.